IDEATION PHASE

💡 Brainstorm & Idea Prioritization Template

📅 DATE 7/9/2026	🧑‍🌾 NAME K.Hemanth Kumar	🌱 PROJECT OptiCrop – Smart Agricultural Production Optimization Engine	🏆 MAX MARKS 4 Marks

🌾 About OptiCrop

OptiCrop is a Machine Learning-powered web application designed to recommend the most suitable crop for cultivation based on soil nutrients and climate conditions. It helps farmers, agribusinesses and agricultural researchers make faster, more reliable, data-driven planting decisions by analyzing soil and environmental readings and determining which crop is most likely to give the best yield. The application evaluates soil and climate patterns using clustering and classification algorithms and deploys the best-performing model through a Flask-based web interface for real-time recommendations.

🎯 Step 1 — Selecting the Problem Statement

👥 Team Gathering

Define who should participate in the session and share relevant information or pre-work ahead.

🎯 Set the Goal

Think about the problem to be solved in the brainstorming session.

❓ Problem Statement

How might we recommend the most suitable crop for a given soil and climate profile using Machine Learning?

🧠 Step 2 — Idea Clusters

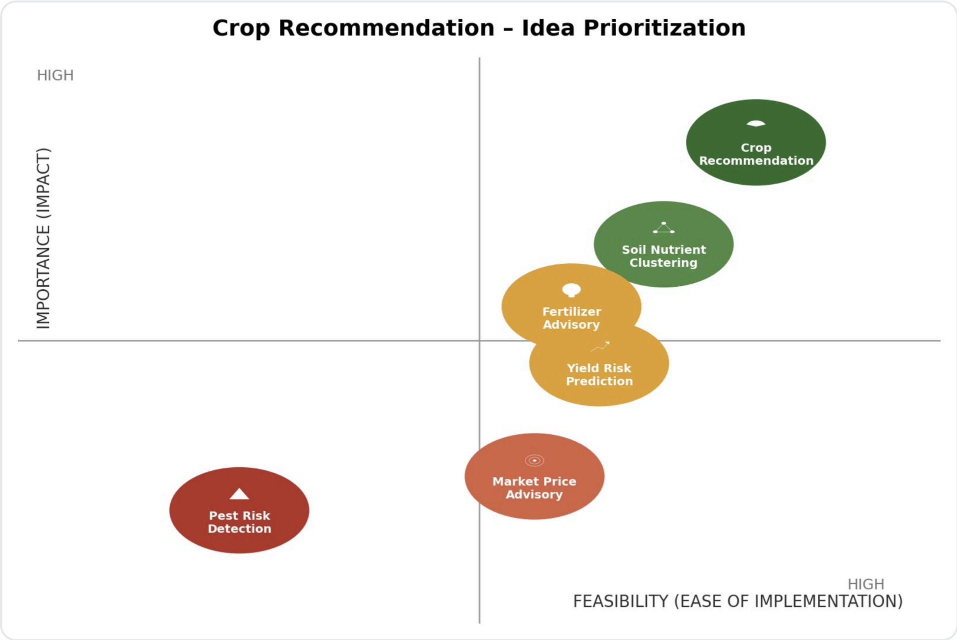
🌿 Idea A – Crop Advisory

Crop Suitability Prediction • Soil Nutrient Analysis • Automated Crop Advisory • Smart Farming Platform • Real-Time Crop Recommendation

💧 Idea B – Resource & Yield Optimization

Fertilizer Requirement Estimation • AI-Based Soil Evaluation • Crop Clustering Analysis • Water Requirement Prediction • Seasonal Crop Planning • Sustainable Farming Support

📊 Figure 1 — Idea Prioritization Quadrant



Idea prioritization quadrant for OptiCrop: Importance (Impact) vs Feasibility

🏅 Step 3 — Idea Prioritization (Importance × Feasibility)

IDEA	IMPORTANCE (IMPACT)	FEASIBILITY	PRIORITY
Crop Recommendation	High	High	1 (build first)
Soil Nutrient Clustering	High	High	2
Fertilizer Advisory	Medium	Medium	3
Yield Risk Prediction	Medium	Medium	4
Market Price Advisory	Low	Medium	5
Pest Risk Detection	Low	Low	6

🚀 Outcome

Faster & accurate crop recommendations → Better yields, lower risk, happier farmers.